

Body systems

THE HUMAN BODY SYSTEMS THAT PERFORM SPECIFIC JOBS.

The human body takes between 18 and 23 years to develop to full size. Medical science divides the body into several body systems—each featuring a set of organs that work together to perform certain jobs.

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Skeletal and muscular systems

Bones give strength and support to the body, and are the main tissue in the skeleton. An adult human skeleton is made up of about 200 bones. These are covered by about 640 skeletal muscles, each one connected by a stiff, cordlike tendon to a specific joint. As the muscle contracts (tightens), it pulls on that joint, creating movement in a variety of ways. Bones are joined to each other by bands of cartilage, called ligaments.

▷ Synovial joints

Most joints are synovial joints—the bone ends have a covering of smooth cartilage and the space between them contains lubricating synovial fluid. Different kinds of joints allow different types of movement.



Ellipsoidal

The wrist has an oval bone sitting in a socket, allowing it to move in two planes—up-and-down, and side-to-side.



Gliding

Gliding joints occur in many places in the skeleton, and are usually very small. They feature bones that are almost flat that can glide over each other.



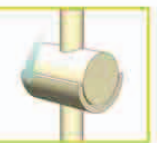
Pivot

The head rotates from side to side using a pivot joint in the neck.



Ball-and-socket

The shoulder can move in all directions thanks to a circular bone connected to a round socket.



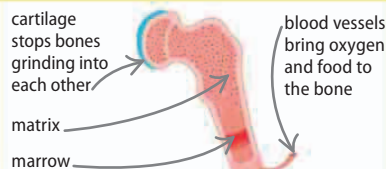
Hinge

Like the hinge on a door, the elbow can move only in one plane—it cannot twist like other joints.



Saddle

Made of two curved bones, a saddle joint allows the thumb to move in two planes.



△ Bone structure

Bones are made of living cells that secrete a matrix of flexible calcium phosphate. In the core, or marrow, of a bone red blood cells are manufactured.

▽ Muscular system

There are two main sets of muscles in the human body. The skeletal muscles work in pairs to move the body, while smooth muscles produce rippling pulses in the digestive system and arteries, to push material along tubes.

